



Facebook to stop political ads

Facebook has said it will stop running all political and social-issue adverts after the election. <http://digit.in/nov20-10>



Tesla's return policy removed

Tesla's 'no questions asked' seven-day return policy has been quietly removed. <http://digit.in/nov20-09>

CUTTING THE CORD

The technological advancements through history that led to the birth and development of wireless technologies

Dhriti Datta | dhriti@digit.in

Right from the inception of telecommunications to the modern era of cellular communications, wireless local networks, and IoT devices, wireless technology has evolved immensely over time. While the past few years have ushered lightspeed innovation and development in this area, the progress wasn't always this speedy. Theories sparked, evolved, materialised and were then commercialised, and all of this happened in what we would deem as

snail's pace in today's dynamic technological landscape. Now, due to the ludicrous advancements in wireless tech, one can share pictures or data in a jiffy over Wi-Fi networks, data or Bluetooth connectivity. Today, we're slowly starting to see the roll-out of 5G networks around the world, capable of data transmission speeds that would be downright unimaginable back during the inception days of wireless technology and communications. Nevertheless, those innovations are what brought us here to the present day, so let's look back at the history of wireless technologies just to appreciate how far we've come today.

THE EARLY DAYS OF WIRELESS COMMUNICATION AND TECHNOLOGY

The very first wireless networks were developed way back, in the pre-industrial era where

people used systems to transmit messages over line-of-sight distances. This would be accomplished by using smoke signals, torch signals, flashing mirrors, signal flares, and semaphore flags. Back then, elaborate signal combinations were used to convey complex information and observation stations were set up on hilltops to relay these messages over long distances. Pigeons were used to deliver messages over long distances, and in some cases, the information was even sent down a stream of water. These archaic forms of communication were later replaced by more technologically-enabled ones.

The origins of wireless technology can be traced back to the year 1865 when a paper about electric and magnetic fields called 'A Dynamic Theory of Electromagnetic Field' was published by Scottish scientists James Clerk Maxwell. This study is often heralded as the work of physics that laid down the foundation for wireless communication. Maxwell theorised that electromagnetic waves could travel at the speed of light and published a set of equations in 1873. Soon after, scientists began putting these equations to practice. Heinrich Hertz proved the existence of electromagnetic waves in the late 1880s and for his work in the field, the international unit used for frequency in radio waves was named after him – Hertz (Hz). The flurry of innovation continued and in the year 1896, the wireless telegraph was introduced.

Guglielmo Marconi, an Italian innovator, began developing a wireless telegraph system utilising radio waves, based on Heinrich's proof of existence, in 1894. Marconi developed a functional



James Clerk Maxwell formulated the theory of electromagnetic radiation

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